HS01: Mapping Molecular Landscapes

Open-Source Approaches to Chemical Space Exploration

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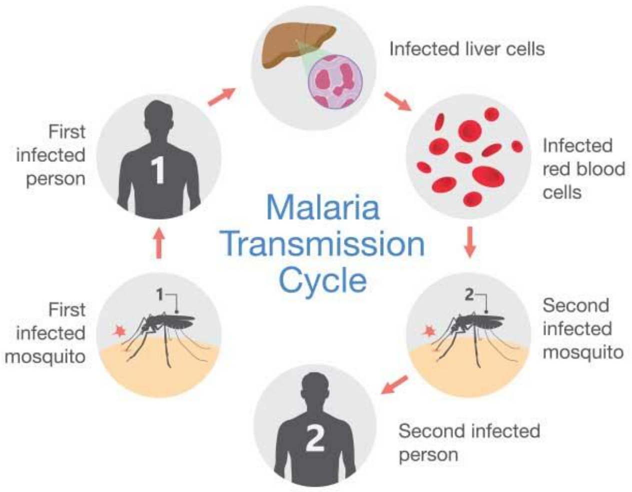
CAiSMD: Computational Applications in Secondary Metabolite Discovery

Malaria To Claim One Million Lives

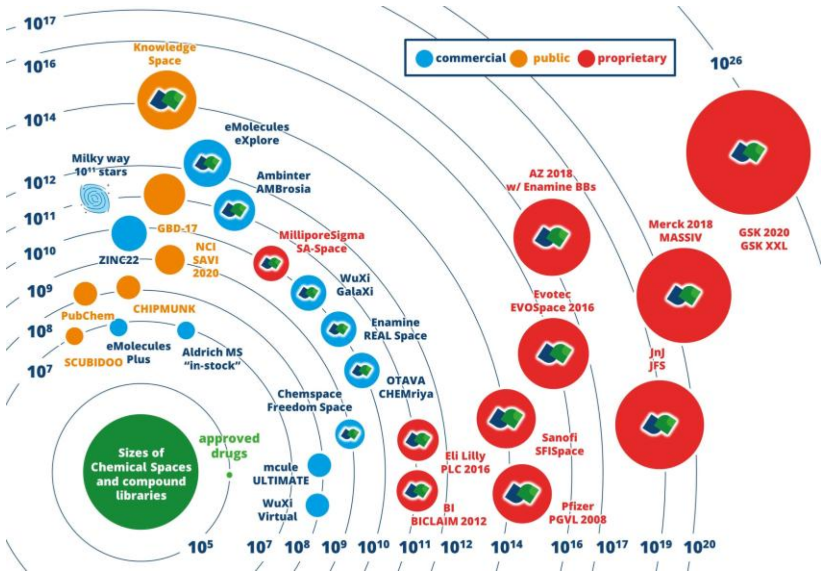
- Malaria affected **249 MM** people in 2023
  - Claiming **569,000** lives
- Its resurgence could trigger **1MM deaths by 2030**
  - Including 750K children under 5 y/o.
- **35-40 approved drugs**, including:
  - Natural products: Quinine, Artemisinin
  - Semi-synthetics: Artesunate
  - Purely synthetics: Chloroquine
- Yet **drug resistance** is rapidly emerging
  - Developing new drugs is critical
  - Funding for neglected diseases is scarce
  - Efficiently navigating the un(der)exploited chemical space is more than every crucial



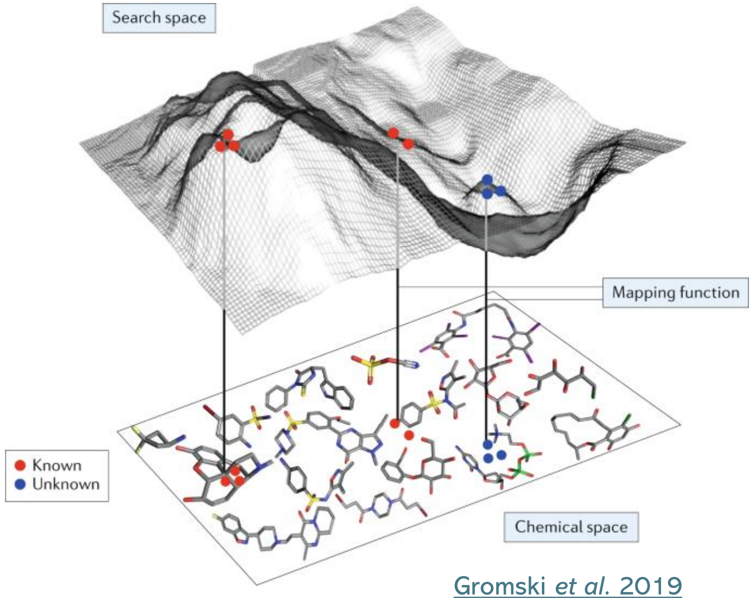
Health Policy Watch, 2025

What is the Chemical Space – Why does it Matter?

- A multi-dimensional manifold encompassing all possible small molecules
  - 10^{60} expected drug-like molecules
  - Only 10^{8} ever synthesized
  - Only ca. 3,500 approved pharmaceutical drugs
- Mapping it is critical for **Structure-Activity Relationship** (SAR) analysis
  - Helps focusing on the "sub-space" most likely to yield viable candidates.
- Navigating it provides the framework for **Virtual Screening** and **Generative AI**.
  - To discover unexploited regions
  - To rapidly find novel, sustainable drugs



Biosolveit.de



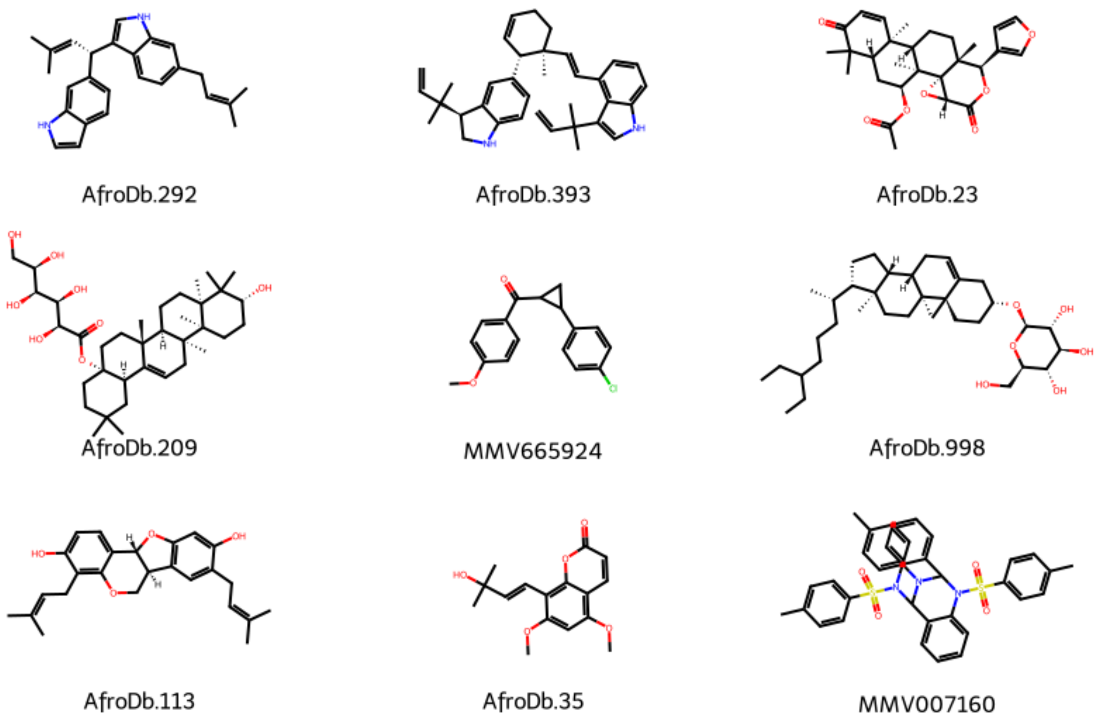
Gromski *et al.* 2019

Learning Objectives

1. Load and Preprocess Molecular Datasets
2. Represent Molecules Numerically
3. Measure Molecular Similarity
4. Reduce and Visualize High Dimensional Chemical Data
5. Interpret Chemical-Space Maps
6. Contextualize Results Biologically

Tools include

RDKIT, Jupyter, UMAP, Plotly, Seaborn, Scikit-Learn



MMV Medicines for Malaria Venture — The Malaria Box

400 structurally compounds with proven anti-malarial activity

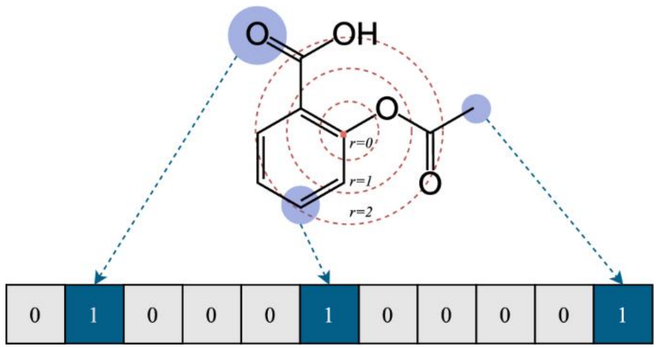
AfroDB: 903 natural products from African medicinal plants by Ntie-Kang et *al.* 2013

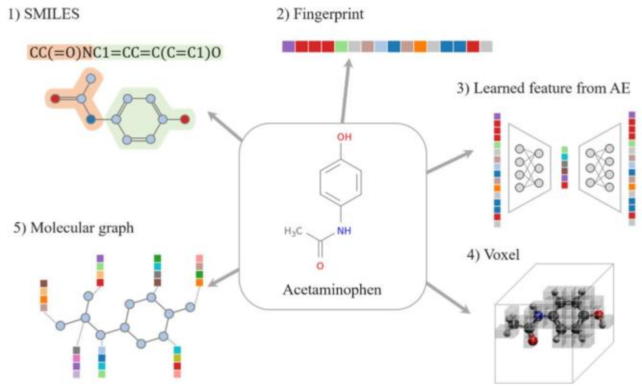
PubChem AID 2302: 13,533 with reported assay data against DD2 provided by GSK

From molecules to Numbers

Morgan Fingerprints

- Iteratively hash each atom's circular neighborhood (radius 2 – ECFP4)
- Return a fixed-length bit vector – each bit marking the presence of a substructure

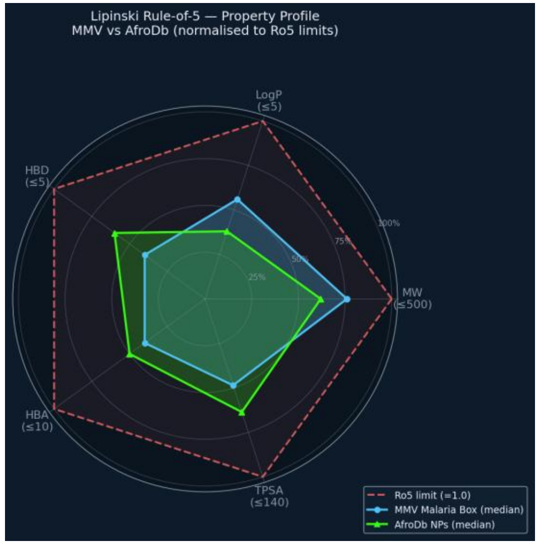




Lipinski Physicochemical Descriptors — ADME representation

- Weight, lipophilicity, topological and conformational descriptors

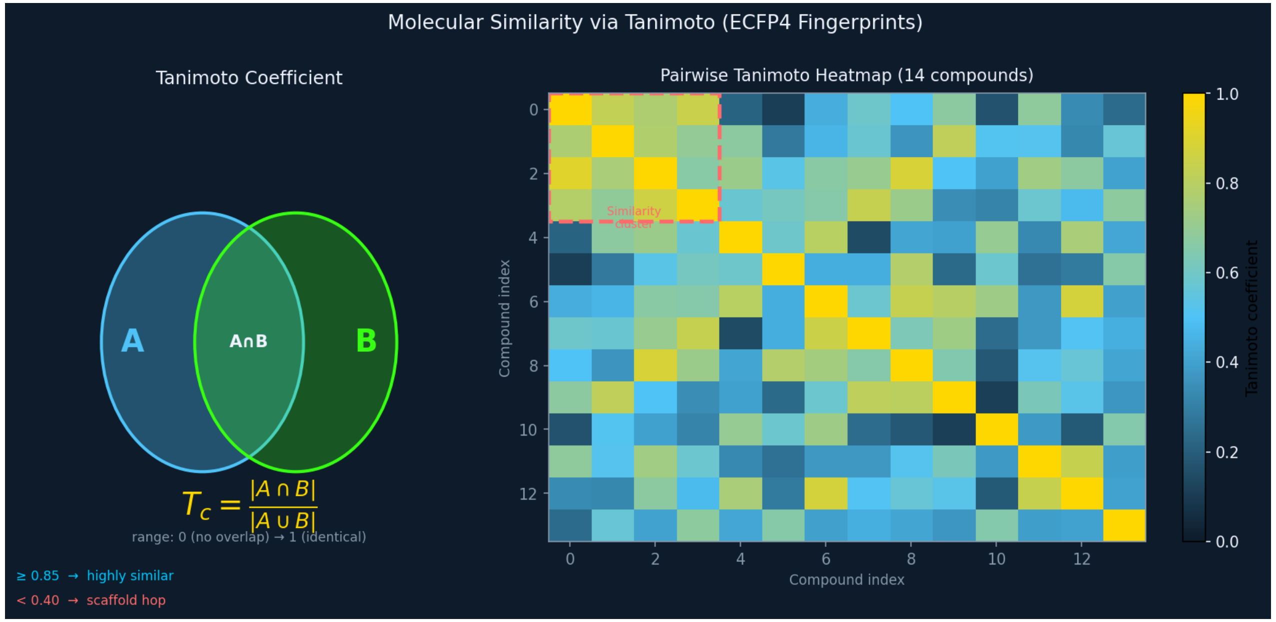
- The are known to significantly impact a molecule’s fate and activity



Measuring Molecular Similarity

Tanimoto Coefficient

- A statistical measure used to quantify the similarity between two molecules
- The value scales from 0.0 (no common fragments) to 1.0 (identical fingerprints).



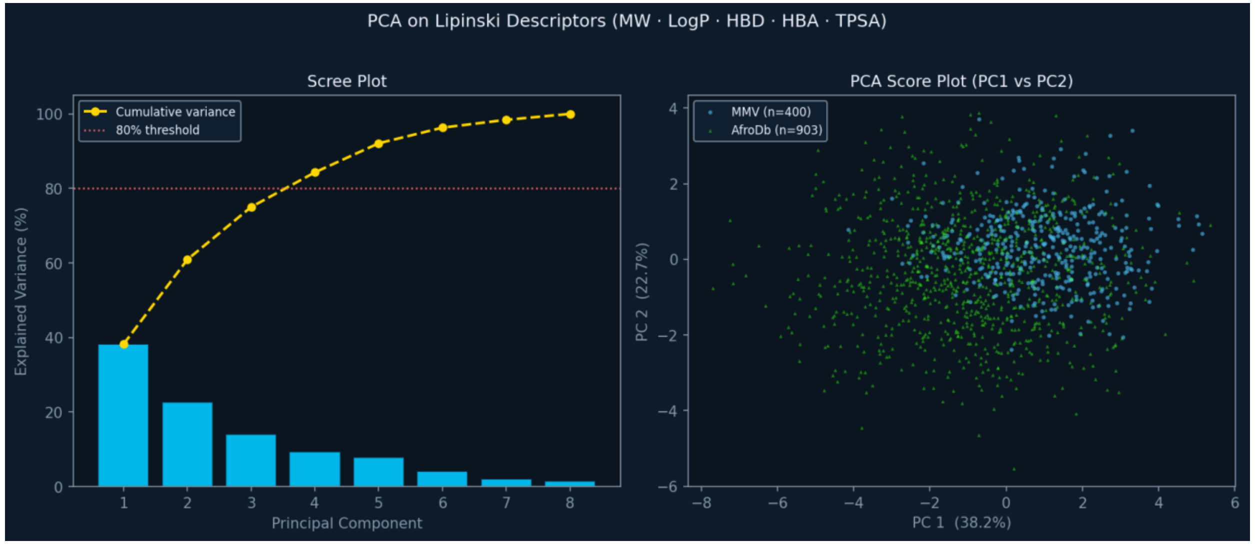
Rules of Thumb

- $T_c > 0.85$: same scaffold – structural analogue
- $0.4 \le T_c \le 0.85$: related chemical class
- $T_c < 0.4$: different chemical class

Dimensionality Reduction and Visualization

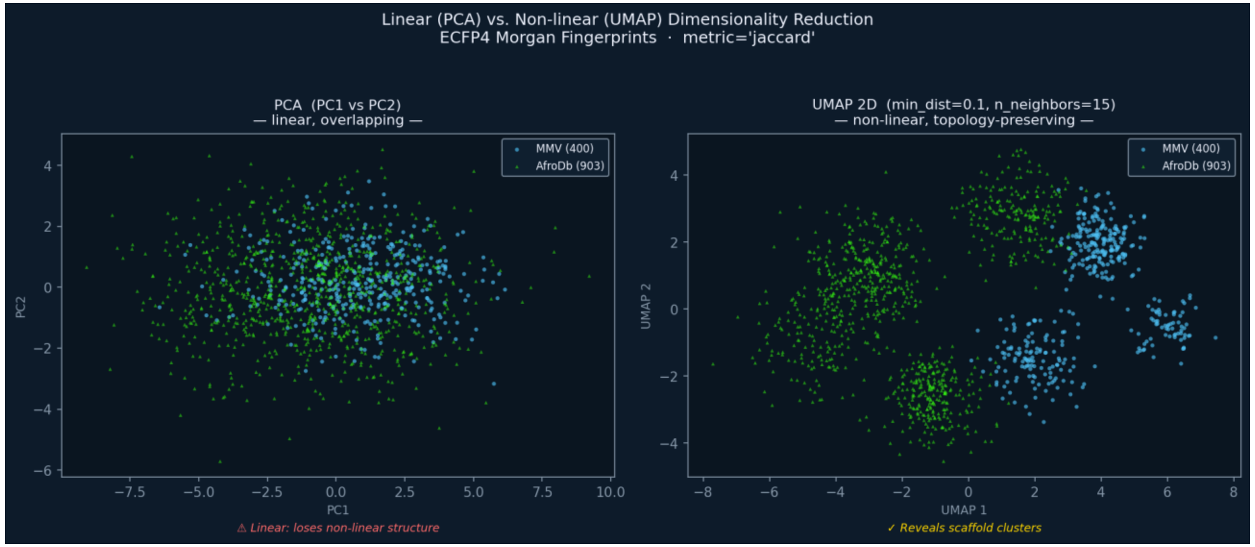
Principal Component Analysis

- Finds the directions of maximum variance in the feature space
- Project onto 2D/3D space
- We will use Lipinski descriptors here



Uniform Manifold Approximation and Projection

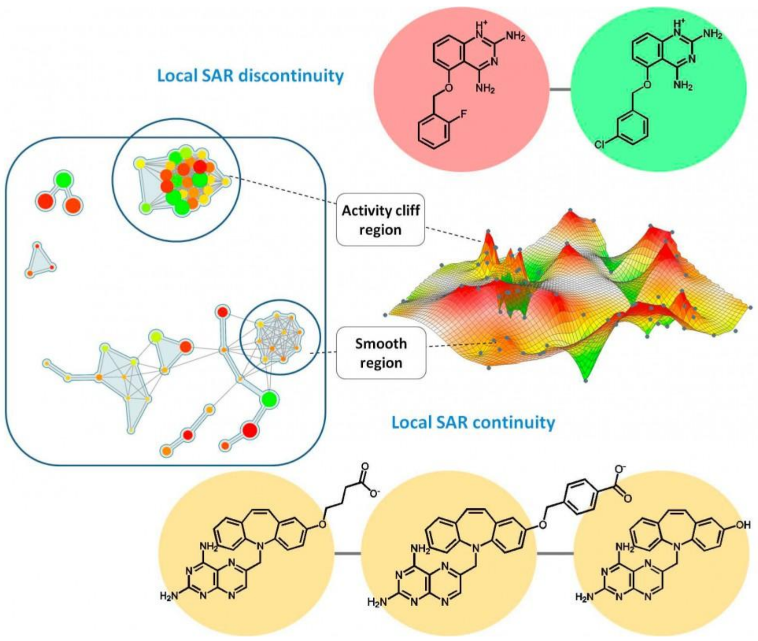
- Builds a weighted k-nearest-neighbor graph
- Similar scaffolds cluster together
- Reveals multiple distinct scaffold clusters — much more biologically interpretable
- We will use ECFP4 fingerprints



These are complementary, not competing. Two molecules can be physico-chemically similar (PCA overlap) yet structurally very different (UMAP separation).

Activity Cliffs

- **Activity cliffs:** two almost-identical molecules, 250× different potency
- Activity cliffs represent "SAR discontinuities"
- They are important in:
  - Lead optimization
  - Bioisostere validation
  - Predictive model failure
- In our context, the help pinpoint the **critical pharmacophoric features** required for the inhibition of key malarial targets



Lab & more

Putting it All Together – Your Workshop Pipeline

Part B Parse SMILES & Salt Removal → **Parts C-D** Fingerprints & Descriptors → **Part E** Tanimoto Similarity → **Part F-G** PCA + UMAP 2D & 3D → **Parts H-I** Property Maps & AfroDb Overlay → **Ext 1–4** Extensions (optional)

- Follow the setup installation in README.md
- Work through the parts in order
  - Fill the blanks as instructed in the comments with each cell (Look for **TODO**)
  - Answer the Stop-and-Discuss questions
  - Tackle as many Extensions as you have time for.
- The notebook is in your repo:
  - https://github.com/djoy4stem/caismd_2026_chemspeace_xplr
  - notebooks/caismd_2026_chemspace_xplr.ipynb.

Extensions & Supplementary Material

Ext 1 — Murcko Scaffold Analysis

Extract Murcko scaffolds from every molecule. Color the UMAP by the top-10 scaffolds. Compute scaffold diversity ratio and identify scaffold-biased regions of chemical space.

Ext 2 — Interactive Molecular Browser

Use mols2grid to browse all 1,303 molecules as a paginated 2D-structure grid. Filter by SMARTS substructure, sort by pIC50 or LogP.

Ext 3 — Activity Cliffs (PubChem AID 2302)

Load a 2,000-compound *P. falciparum* screen. Find all pairs with Tanimoto ≥ 0.9 and opposite activity labels. Draw the top-5 cliff pairs side-by-side.

Ext 4 — AfroDb Projected onto Drug Space

Project AfroDb onto the MMV-only UMAP. Identify which African NPs land nearest to known active clusters — a virtual screening shortlist.

Extension J — Functional Group Profiling (separate notebook)

Deeper interpretable fingerprinting: MACCS keys · Ertl functional groups · chi^2 enrichment test · Butina clustering · MACCS vs ECFP4 UMAP comparison. ~65 min standalone session.

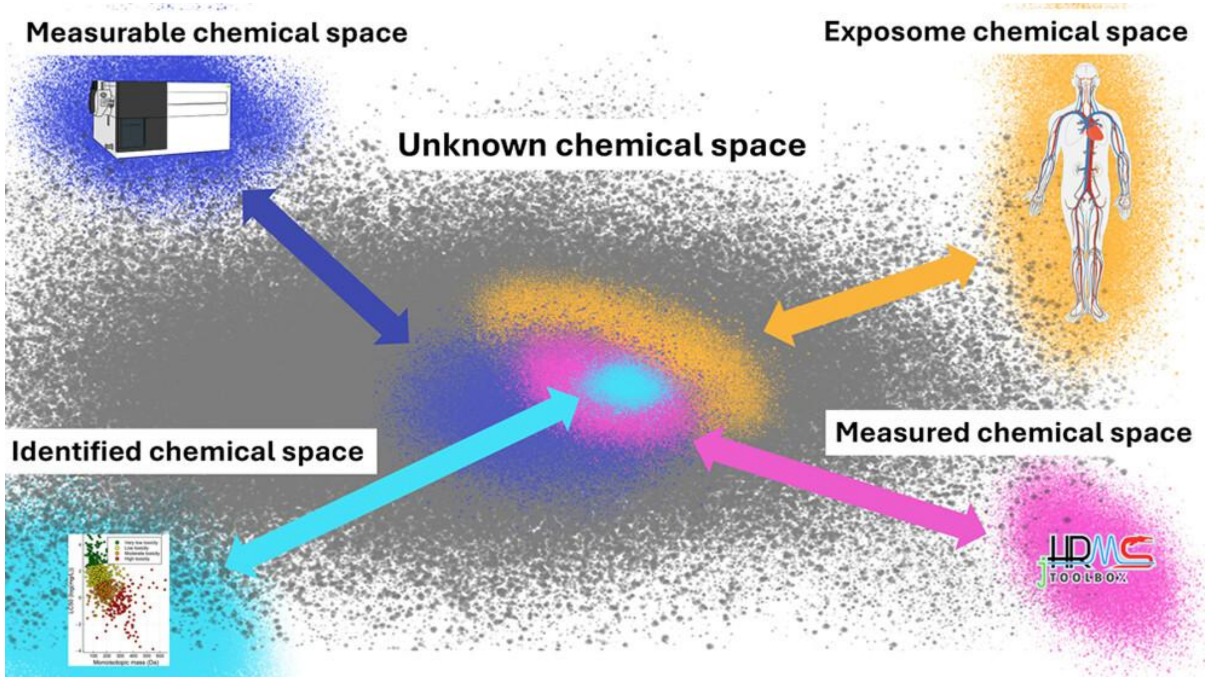
Examples of Interactive Graphical Tools

Tool	Best For...	Technical Level	Key Feature
DataWarrior	Rapid SAR exploration & plotting	Beginner (GUI)	Standalone desktop app; built-in property calculators.
MAUD	Interactive activity-space mapping	Beginner (Web)	Seamlessly overlays potency/activity data onto structural clusters.
RDKit + UMAP	Custom pipelines & large data filtering	Intermediate (Python)	High flexibility; easy to integrate into Jupyter notebooks.
TMAP	Visualizing massive datasets (100k+)	Intermediate (Python)	Uses "Tree-MAP" to show hierarchical structural relationships.
ChemPlot	Quick visual clusters for reports	Beginner/Int.	Automated generation of UMAP/t-SNE/PCA from SMILES.
Maya	Multi-objective SAR	Beginner (Web)	Visualizing multi-target activity & selectivity

Thank You

- The organizing committee
- The listeners

Let's Have Some Fun!



Samanipour *et al.* 2024